Questions-Gen — Math competition problem generator (Qwen3-14B)

github.com/chenxingqiang/GenQuestion · Unsloth training · Hugging Face: xingqiang/* · pip install questions-gen

1. Training mixture

Historical contests
50% · CUMCM / olympiad

Conditional variants
30% · rewrite constraints

Innovative types
20% · new stems

data_preparer.py
tokenize / pack

2. Three-stage pipeline

Stage 1 — Pretrain
Qwen3-14B SFT
HF: stage1-fp-merged
base generation

Stage 2 — GRPO
8 problems / group
novelty constraint
policy optimization

score

Reward
0.4 difficulty
0.3 novelty
0.2 rigor
0.1 diversity

eval

Stage 3 — Distill
DeepSeek-R1 teacher
difficulty / logic / edu
final merged FP16

3. Validation (questions_gen/validation)

ModelValidator
single-model n tests

BatchValidator
algebra / calculus
geometry / number theory

QualityEvaluator
content / clarity
difficulty / originality

Scores
Final 0.847
teacher 4.2 / 5

4. Package / deploy

CLI
questions-gen
validate / batch
quality / compare

Python API
QuestionsGenTrainer
OllamaManager
HuggingFaceUtils

Hugging Face
stage1 / stage2 / final
xingqiang/*

Ollama
push-all
local generate API

Pretrain → GRPO with multi-dimensional rewards → R1 distillation → validate by category → publish HF weights and Ollama models.